

PartB_Interim_DRAFT_for_Turnitin.pdf

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Diabetic Retinopathy Severity Grading: A Controlled Comparison of Four Architecture Families

ICT 4442 Deep Learning Mini Project, Interim Report (Part B)

Manipal Institute of Technology

1. Literature Review

Ten papers. Three set the clinical baseline for deep learning on diabetic retinopathy, five cover the architecture families our four models are drawn from, one gives the ordinal machinery our output heads use, and one defines our headline metric.

1.1 Summary table

Paper (Author, Year)	Method	Dataset	Key result	Relevance to project
Cohen, 1968	Weighted kappa: extends chance-corrected agreement to ordered categories by weighting each disagreement by the squared distance between grades, so an error of three counts nine times an error of one	Methodological, no dataset	An agreement statistic that is 1 for perfect prediction and 0 for a predictor carrying no information beyond the marginals	Quadratic weighted kappa is our headline metric and the metric APTOS 2019 itself scored on. We verified the degenerate case against our own label file: predicting No DR every time gives 49.29% accuracy and exactly 0.0 kappa, which

Paper (Author, Year)	Method	Dataset	Key result	Relevance to project
Simonyan and Zisserman, 2015	Depth studied in isolation by stacking only 3x3 convolutions, 16 to 19 weight layers, doubling channel width after each pooling stage	ImageNet ILSVRC-2014	7.3% top-5 test error, first place in localisation and second in classification at ILSVRC-2014	is why accuracy alone would mislead on this dataset Our Model 2 is this design shrunk to five stages of two 3x3 convolutions each, 32 to 512 channels, 4.72M parameters. Their result is why a from-scratch CNN built this way is a fair baseline rather than a strawman
He et al., 2016	Residual learning: layers fit a residual relative to their input via identity skip connections, which keeps gradients usable at depth	ImageNet, CIFAR-10, COCO	3.57% top-5 error on ImageNet with an ensemble, first in ILSVRC-2015 classification. 152 layers, 8x deeper than VGG at lower complexity. 28% relative gain on COCO detection	ResNet-50 is in our model registry as a control, and ResNet-34 is the backbone CORAL was validated on, so it is the bridge between our architecture axis and our head axis
Tan and Le, 2019	Compound scaling: depth, width and input resolution scaled	ImageNet, transferred to CIFAR-100 and Flowers	B7 reaches 84.3% top-1 while being 8.4x smaller	B0 is our Model 3. At 4.01M feature-extractor

Paper (Author, Year)	Method	Dataset	Key result	Relevance to project
Dosovitskiy et al., 2021	together by one coefficient found by grid search, applied to an NAS-designed baseline of inverted-residual MBConv blocks with squeeze-and-excitation		and 6.1x faster than the previous best. B0 reaches 77.1% top-1 with 5.3M parameters (Table 2). Transfers to 91.7% on CIFAR-100 and 98.8% on Flowers	parameters it is smaller than our from-scratch CNN, which is what lets us argue that any gain over that CNN comes from pretraining and not from capacity
	Pure transformer on 16x16 image patches, no convolutions.	JFT-300M and ImageNet-21k	ViT-H/14 reaches 88.55% on ImageNet after JFT-300M pretraining, 90.72% on ImageNet ReaL and 77.63% on VTAB. Trained on ImageNet-1k alone, ViT underperforms ResNets, which the authors attribute to the absence of translation equivariance and locality	ViT-S/16 is our Model 4. Their negative result is the more useful one for us: our training set is 3,662 images, roughly 1/82,000 of JFT-300M, so this paper predicts the ViT should lose to the CNNs here. That prediction is testable and it is what Objective 1 is for
	Patch embedding, positional encoding, standard transformer encoder	for pretraining, transferred to ImageNet, CIFAR, VTAB	ImageNet-1k	
Touvron et al., 2021	DeiT: a ViT trained on ImageNet only, using heavy augmentation, regularisation and a distillation	ImageNet-1k only, no external data	83.1% top-1 with 86M parameters and no external data, up to 85.2% with	The counterweight to the paper above. It shows the ViT data requirement is partly a

Paper (Author, Year)	Method	Dataset	Key result	Relevance to project
	token that lets the transformer learn from a CNN teacher through attention		distillation, trained in under 3 days on one machine	training-recipe problem, not purely architectural, so if our ViT does badly we can separate “wrong architecture for this data scale” from “wrong recipe”. DeiT-S is in our registry as the fallback backbone Our CORAL head is this, with $K = 5$. Weight sharing is the whole point: it is what makes rank monotonicity structural rather than something the optimiser has to be trusted to learn. DR grades are ordered in exactly the sense the paper assumes, and the head axis of our 4x3 ablation is
Cao, Mirjalili and Raschka, 2020	CORAL: an ordinal label is extended into K-1 binary “is the grade above k” tasks. The K-1 classifiers share one weight vector and have independent bias units. Theorem 1 proves the learned biases come out ordered, $b_1 \geq b_2 \geq \dots \geq b_{K-1}$, so the predicted cumulative probabilities are non-increasing by construction	MORPH-2 (55,608 face images), AFAD and CACD, age estimation with ResNet-34	MAE on MORPH-2 falls from 3.34 with plain cross-entropy and 2.83 with the earlier OR-CNN to 2.64. Same direction on AFAD (3.60 / 3.51 / 3.47) and CACD (5.65 / 5.38 / 5.25), Table 2	

Paper (Author, Year)	Method	Dataset	Key result	Relevance to project
Gulshan et al., 2016	Inception-v3 ensemble trained for binary referable DR, each development image graded 3 to 7 times by a panel of 54 US-licensed ophthalmologists and senior residents	128,175 images for development, validated on EyePACS-1 (9,963) and Messidor-2 (1,748)	AUC 0.991 on EyePACS-1 and 0.990 on Messidor-2. At the high- sensitivity operating point, 97.5% sensitivity and 93.4% specificity on EyePACS-1, and 96.1% / 93.9% on Messidor-2	softmax against regression against this The paper that established CNNs work at screening scale, and our scale contrast: 128,175 images against our 3,662, roughly 35x. It also treats referable DR as binary rather than five ordinal grades, which is why we report referable-DR F1 alongside kappa as the bridge between the two framings
Ting et al., 2017	Deep learning system for referable DR, vision- threatening DR, glaucoma and AMD, developed on a national screening programme and validated across multiple ethnicities	494,661 images in total. Primary validation on 71,896 images from 14,880 patients, plus 10 external multiethnic datasets across Singapore, China, the US, Australia,	Referable DR: AUC 0.936, sensitivity 90.5%, specificity 91.6%. Vision- threatening DR: AUC 0.958, sensitivity 100%, specificity 91.1%	The external- validation counterpoint to Gulshan. Performance drops from 0.99 to 0.936 AUC once the test population is not the development population, which is the honest ceiling

Paper (Author, Year)	Method	Dataset	Key result	Relevance to project
		Mexico and Hong Kong		on what a single-source dataset like APTOS can claim. It is the main limitation we state in section 5
Pratt et al., 2016	CNN trained from scratch with data augmentation for all five DR stages, learning lesion- level features (microaneurysms, exudates, haemorrhages) without hand- crafted features	Kaggle EyePACS, 80,000 fundus images, 5,000 held out for validation	95% sensitivity and 75% accuracy on the five-class task	The closest prior work to ours: same five-class ordinal framing, same Kaggle image source, also from scratch. Their 75% accuracy on 80,000 images is the reference point our from-scratch CNN is measured against on 3,662, and the gap tells us how much of their result was data volume

1.2 What the table shows when read across

Three comparisons matter for this project.

Data scale is the variable nobody controls for. Ting used 494,661 images, Gulshan 128,175, Pratt 80,000. We have 3,662. That is between 20 and 135 times

less, and it is not a gap we can close, so it has to be designed around. It is the reason two of our four models are ImageNet-pretrained rather than trained from scratch, and the reason we expect the ViT to be the weakest of the four rather than the strongest. Dosovitskiy et al. and Touvron et al. disagree about how much of that requirement is architectural and how much is a training recipe, and at our data scale that disagreement is directly testable.

The clinical papers and the architecture papers use different targets. Gulshan and Ting both collapse the problem to a binary referable-or-not decision, which is the clinically actionable question but discards the grade ordering. Pratt keeps all five classes but scores with plain accuracy, which treats confusing grade 0 with grade 4 as no worse than confusing grade 3 with grade 4. Cohen's weighted kappa is what makes that distinction, and CORAL is what builds the ordering into the model rather than only into the score. None of the three DR papers uses both. We report quadratic weighted kappa as the headline and referable-DR F1 alongside it, so our results can be read against either literature.

No paper here varies architecture and output head independently. Each of the five architecture papers fixes a task framing and compares backbones. CORAL fixes a backbone, ResNet-34, and compares heads. The gap is the interaction: whether the benefit of an ordinal head depends on which backbone it sits on, and whether it is larger when the backbone is weak. That is what our 4x3 grid of four backbones against three heads, twelve runs on one fixed split with one fixed schedule, is built to answer. It is a small contribution but it is a real one, and it is the reason our objectives are framed as a controlled comparison rather than as an attempt to beat a leaderboard.

1.3 References

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Part B, Section 2: Dataset Acquisition and Preprocessing

Everything below is written against code that is committed and numbers that were measured, not estimated.

2.1 Acquisition

We use the APTOS 2019 Blindness Detection dataset, collected by the Asia Pacific Tele-Ophthalmology Society from Aravind Eye Hospital in Tamil Nadu, India, and released through Kaggle. The download is scripted in `scripts/download_data.sh` so that every member builds the same local copy from the same source.

The competition ships 3,662 labelled training images and 1,928 test images whose labels are withheld for scoring. Because those labels are not public, the test set is useless to us for measuring anything, so every split in this study comes out of the 3,662 labelled images. This is worth stating plainly because papers reporting APTOS numbers are not always clear about which set they scored on.

The raw download is 8.01 GiB. The images are not uniform: there are 17 distinct resolutions, from 474 x 358 up to 4288 x 2848, with the most common single size (1050 x 1050) covering only 26.6% of the set. That spread is the signature of several different fundus cameras used across multiple screening sites, and it is the first thing preprocessing has to remove.

Grades follow the International Clinical Diabetic Retinopathy Severity Scale: 0 No DR, 1 Mild, 2 Moderate, 3 Severe, 4 Proliferative. The distribution is 1805 / 370 / 999 / 193 / 295, a 9.4:1 ratio between the largest and smallest class.

2.2 The preprocessing pipeline

Implemented in `src/preprocess.py`, committed as `6dfcbf0`. One command builds the cache:

```
python -m src.preprocess --raw data/raw/aptos2019 --out data/processed --s
```

Four stages, applied to every image in the same order. Figure 1 walks a single grade 3 image through all of them.

Cropping the fundus circle. A fundus photograph is a circle of retina on a black rectangle, and the amount of black differs by camera. `crop_fundus` converts to greyscale, thresholds at intensity 7, and trims to the bounding box of what survives. Thresholding on greyscale rather than per channel matters here: the retina is strongly red dominant, so a per channel threshold keeps border pixels that are dark in red but not quite zero in blue, and the crop comes out loose. If every pixel falls below the threshold the original image is returned unchanged rather than reduced to nothing.

Resizing. Cropped images go to 320 x 320 with `INTER_AREA`, which averages over the source pixels it is collapsing. Since almost every image in this set is being shrunk, and by a large factor, area interpolation avoids the aliasing that nearest neighbour or bilinear sampling would introduce on fine vessel structure.

Illumination normalisation. This is the Ben Graham method from the winning entry of the 2015 Kaggle Diabetic Retinopathy competition, and it is still the default a decade later. Each image is blended against a heavily blurred copy of itself:

```
output = 4 * image - 4 * GaussianBlur(image, sigma) + 128
```

with `sigma` set to `size/30`, which is 10.67 px at our 320 px working size. The blurred copy holds the low spatial frequency content, which is almost entirely the illumination gradient across the retina and the overall colour cast of the camera. Subtracting it leaves the high frequency content, which is where the lesions are: microaneurysms, haemorrhages, hard exudates. The effect is that two photographs of the same eye taken on different machines end up looking far more alike than they did, without the lesions themselves being touched. The 128 offset re-centres the result into the middle of the 8 bit range.

Circular masking. Normalisation leaves a bright halo where the Gaussian blur straddles the boundary between retina and black surround, visible in the fourth

panel of Figure 1, and cropping leaves four dark corners outside the circle. `circle_mask` zeroes everything beyond 0.95 of the half-width, so the network cannot pick up on corner artefacts that correlate with camera model rather than with disease. Measured over every twentieth cached image, 29.17% of pixels are zero, against 29.12% predicted analytically for a circle of that radius, which confirms the mask is landing where it should.

Caching. Results are written as a single `uint8` array, `images_320.npy`, of shape (3662, 320, 320, 3) and 1073 MiB, with `labels.csv` written in the same row order. Figure 2 shows five images from five different cameras before and after the pipeline: the raw row runs from pale grey-pink through saturated orange to dark olive, and the processed row is close to a common grey-blue with the exudates standing out as bright yellow in grades 3 and 4. That is the whole point of the normalisation, and it is easier to see than to argue.

Caching as `uint8` rather than `float32` costs a quarter of the memory and lets the whole set sit in page cache during training, which is what makes 12 training runs practical on one machine. Conversion to float and channel normalisation happen per batch in the dataset class instead.

The job is parallelised across processes, defaulting to `cpu_count() - 2` workers. OpenCV's own threading is switched off inside the pool, since otherwise each worker spawns its own threads and they contend for the same cores. On 8 workers the full set takes 2 minutes 15 seconds.

2.3 Figures

Generated by `scripts/make_figures.py`, so they rebuild from the raw data rather than being pasted in from a notebook:

- **Figure 1**, `fig1_pipeline_stages.png`. One image at each stage: raw 3216 x 2136, cropped to 2709 x 2136, resized to 320, normalised, masked.
- **Figure 2**, `fig2_camera_normalisation.png`. Five cameras, one per distinct raw resolution, spanning all five grades, raw against processed.
- **Figure 3**, `fig3_grade_distribution.png`. Grade counts across the 3,662 cached images.

2.4 Verification

The pipeline was checked rather than assumed:

- All 3,662 images decoded. Nothing was dropped as unreadable.
- The class distribution of the cache is 1805 / 370 / 999 / 193 / 295, matching the raw `train.csv` exactly.
- The `id_code` order in `labels.csv` matches the raw `train.csv` order.

- Ten sampled rows, including the first and the last, are pixel identical to a serial recomputation of the same images. This is the check that matters most, because the pool returns results out of order; writing them to the wrong index would silently pair every image with someone else's label and still produce a cache that looks fine.
- No cached image is entirely black.

2.5 Augmentation

Augmentation is applied on the fly inside `APTOSDataset` and only when the dataset is constructed with `train=True`. Validation and test see the cached image unchanged, so the reported metrics are never measured on a transformed image.

Transform	Parameter	Probability
Horizontal flip	axis 1	0.5
Vertical flip	axis 0	0.5
Rotation	uniform on [-180, 180] degrees	always
Scale	uniform on [0.9, 1.1], applied in the same affine warp	always
Brightness gain	uniform on [0.9, 1.1] multiplicative	always
Brightness offset	uniform on [-10, 10] on the 0-255 scale	always

The geometric half of this is more aggressive than would be sensible for natural images, and deliberately so. A fundus photograph has no canonical up: the retina is imaged through a circular aperture and the camera can be rotated freely at capture time without changing what a clinician would grade. Both flips and full 360 degree rotation are therefore label preserving here in a way they would not be for, say, street scenes. Rotation and scale are composed into a single `cv2.warpAffine` so the image is resampled once rather than twice, with black used for the corners exposed by rotation, which matches the black background already outside the retinal circle.

The photometric half is deliberately mild. Ben Graham normalisation has already removed most of the inter-camera colour and illumination variation, which is the variation colour jitter would normally be introduced to simulate. Adding strong jitter on top would be undoing preprocessing we ran on purpose, and the lesions that separate grade 1 from grade 2, microaneurysms a few pixels across, are exactly the low-contrast structures that heavy photometric noise destroys. A gain

of plus or minus 10 percent and an offset of plus or minus 10 grey levels is enough to stop the network keying on absolute intensity without erasing that signal.

2.6 Train / validation / test split

The split is stratified by grade, drawn once with seed 42, and cached to `data/processed/split_seed42.json` as three lists of row indices. Every run in the comparison loads that file rather than re-drawing, so all twelve backbone-by-head combinations see exactly the same three sets. The proportions are 70 / 15 / 15.

Grade	Full	Train	Val	Test
0, No DR	1805	1263	271	271
1, Mild	370	259	55	56
2, Moderate	999	699	150	150
3, Severe	193	135	29	29
4, Proliferative	295	207	44	44
Total	3662	2563	549	550

Stratification holds to three decimal places in all three splits. Grade 0 is 49.29 percent of the full dataset and 49.28, 49.36 and 49.27 percent of train, validation and test respectively. Grade 3, the rarest at 5.27 percent, gives 29 images in each of validation and test, which is small enough that per-grade recall on that class will be noisy and we say so in the results rather than reading much into single-image differences.

We verified the three sets are a genuine partition: disjoint pairwise, exhaustive over all 3,662 rows, and reproducible from the seed. Stratifying matters more here than it would on a balanced dataset. Drawn at random without it, a 15 percent test split would have a standard deviation of about five images on grade 3 alone, which is enough to move quadratic weighted kappa by more than the differences we are trying to measure between models.

2.7 Correction to the Synopsis

The synopsis gave the maximum image size as 3216 x 2136. Measured across all 3,662 files, the true maximum is 4288 x 2848. The figure is corrected here and should be carried into the final report.

Part B, Section 5: Risk and Plan for Remaining Work

5.1 Where the project stands

Committed and working: the preprocessing pipeline (`src/preprocess.py`), the shared evaluation protocol (`src/metrics.py`), the two pretrained backbone wrappers (`src/models/pretrained.py`), and the common training harness (`src/train.py`). Five commits, per member, on a shared repository.

Outstanding: the dataset class and fixed split, the MLP, the CNN, the three output heads and the loss module.

5.2 Risks

The critical path runs through two files, not through the models. Nothing in this project can train until `src/data.py` (`make_split`, `APTOSDataset`) and `src/models/heads.py` plus `src/losses.py` exist. Every model file already takes a tensor and returns a tensor, so the four models can be written and shape tested in parallel today, but not one of them can produce a number until those two pieces land. They are due 3 September and 8 September respectively. This is the single largest risk to the 25 September interim deadline and it is a scheduling risk rather than a technical one.

Mitigation. Both are small, well specified files with the input and output contract already written into their docstrings. If either slips past 8 September, the owner hands over what exists and a second member finishes it, with the handover recorded in the contribution log rather than hidden.

Not every member has a GPU. We measured this rather than guessing. On a 6 GB RTX 3060, throughput is 35,700 img/s for the MLP, 419.6 for EfficientNet-B0, 354.1 for the from-scratch CNN and 339.6 for ViT-S/16. On 8 CPU threads the same models run at 1166.5, 7.8, 7.3 and 7.6 img/s. That is roughly a 50x gap, and it puts a full training run at 3 to 5 hours on CPU against a few minutes on the GPU.

Mitigation. CPU training is slow but not blocking, since a 3 to 5 hour run finishes overnight, and the harness already runs correctly without CUDA. For members who would rather not tie up a laptop, Kaggle Notebooks give 30 GPU hours a week on a P100, with APTOS already mounted, which is more than the whole comparison needs. Preliminary results for the interim report do not have to come from the fastest machine.

Twelve runs on one 6 GB card. The full study is four backbones times three heads. Measured at batch 32 with mixed precision (256 for the MLP), no model exceeds 1.5 GB, so the memory budget is comfortable. Wall clock is the real constraint rather than VRAM.

Mitigation. The interim report needs only two models with preliminary, explicitly unoptimised results. The remaining ten runs are scheduled against milestone 6 on 14 October, which leaves seventeen days of slack before the final deadline.

ViT-S/16 may simply underperform on 3,662 images. Vision transformers are data hungry, and this is a small dataset by their standards. A weak ViT number is a legitimate finding for Objective 1 rather than a failure, but a ViT that will not converge at all is a problem.

Mitigation. `deit_small_patch16_224` is already wired into the registry as a drop-in fallback. It is the same size as ViT-S but was trained with much heavier augmentation, which usually transfers better to small datasets. Similarly `resnet50` stands behind EfficientNet-B0 if that proves unstable.

Class imbalance at 9.4:1. A model that answers No DR every time scores 49.3% accuracy. Reporting accuracy alone would flatter every model in the study.

Mitigation. Already handled in the evaluation protocol. Quadratic weighted kappa leads the comparison table, and that same all-No-DR model scores exactly 0 on it, which we verified numerically against the real label file. Macro F1 and referable-DR F1 are reported alongside. Inverse frequency class weighting is available behind a `--balanced` flag so its effect can be measured rather than assumed.

Mid-term examinations fall on 1 October, six days after the interim deadline. The last week of September is not usable for project work.

Mitigation. Interim content is due internally on 20 September, five days before submission, so that the 25 September deadline is an assembly job rather than a writing job.

5.3 Timeline to final submission

Date	Work	Owner
03 Sep	<code>data.py: make_split,</code> <code>APTOSDataset</code>	Aditya
08 Sep	<code>heads.py, losses.py;</code> harness runs end to end	Saurabh
20 Sep	Models 1 and 2 with preliminary results; literature review table	Aditya, Harshit; all

Date	Work	Owner
25 Sep	Interim report, Part B	all
14 Oct	Models 3 and 4 to convergence; full 4 x 3 ablation	Vishrut, Saurabh
26 Oct	Error analysis, comparison table, draft report, Turnitin	Vishrut; all
26-29 Oct	Presentation and individual viva	all
31 Oct	Final report and repository	all

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